Hari Prakash

ee22resch01001@iith.ac.in | hari.prakash29695@gmail.com | +91 9566258045 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	9.00
Srimathi Maharanibai Jumna Doss Vaishnav Higher Secondary School	Class XII (State board)	2013	74.25%
Kola Perumal Chetty Vaishnav Senior Secondary School	Class X (CBSE)	2011	6.00%

SCHOLASTIC ACHIEVEMENTS

National Winner, Tata Steel Materials NEXT 3.0 Open Innovation Challenge (2022), Team Nano Tribe: Self-powered sensor for detecting pre-diabetes through exhaled human breath.

Granted Indian Patent: In situ preparation of high-purity copper-copper oxide nanocomposite powder for supercapacitor applications; **Patent No. 590359**, granted **26 May 2026**.

Published Indian Patent Application: Turmeric-biomass-derived carbon electrodes for dual-carbon sodium-ion capacitors; **Application No. 202641066480**, published **5 June 2026**.

Three additional Indian patent applications published:

1. An improved environment-friendly process for the synthesis of high-purity vanadium oxide-graphene composite nanopowder, (**202421090629**);
2. Method for the publication of heavily aluminum-doped zinc oxide nanopowder at room temperature (**202521061468**);
3. Electrochemical Oxidation for the synthesis of Titanium Dioxide Nanopowder (**202521094860**).

Co-authored two papers in the Journal of Energy Storage:

1. Revealing the supercapacitive performance of N-doped hierarchical porous activated carbon in aqueous, ionic liquid, and redox additive electrolytes (**DOI: 10.1016/j.est.2022.105189**)
2. Facile synthesis of ZnTiO3 nanoflakes as an efficient electrode material for high energy density supercapacitor applications (**DOI: 10.1016/j.est.2022.106114**).

First-author manuscript under review, Journal of Energy Storage: "Sustainable Dual-Carbon Sodium-Ion Capacitor from Turmeric Biomass Derived Carbon Without Doping or Electrolyte Additives: A Pouch Cell-Level Validation."

PROJECTS

Biomass-Derived Carbon for Dual-Carbon Sodium-Ion Capacitors | Ph.D., IIT Hyderabad

- Developing biomass-derived carbon electrodes (activated carbon and hard carbon) for dual-carbon sodium-ion capacitors.
- Fabricated pouch cells and evaluated electrochemical performance to validate the electrode materials at the device level.

3D-Printed Triboelectric Nanogenerators | Ph.D., IIT Hyderabad

- Acquired skills in triboelectric nanogenerator synthesis and fabrication within **three months**, then adapted previous modification designs to build a customized 3D-printing setup.
- Prepared materials and slurries, fabricated devices, and measured electrical output using the modified setup with microstepping adjustments.

3D-Printed Supercapacitors | Master's Thesis

- Synthesized few-layer graphene oxide, formulated electrode slurries, and fabricated and tested supercapacitor devices through syringe-based paste extrusion.
- Designed a custom syringe extruder in Autodesk Fusion 360 and used the existing printer to manufacture its modification parts.
- Modified and flashed Marlin firmware to configure the printer for syringe-based extrusion in place of filament

deposition.

POSITIONS OF RESPONSIBILITY

- **Product Design and Development Manager, three-person face-shield venture:** Designed products in Autodesk Fusion 360, resolving printability issues and simplifying fabrication before production.
- Coordinated outsourced manufacturing with 3D-printing providers and injection-molding companies during the first COVID-19 wave; the venture sold **50,000 face shields within 45 days** and was featured in **The Hindu**.
- Secured a single order for **10,000 face shields valued at INR 7.5 lakh** during six months of marketing involvement.
- Delivered **200+ hours of lectures over four years** to B.Tech., M.Tech., and Ph.D. students.

SKILLS

- **Materials synthesis:** Wet- and dry-processing techniques, hydrothermal synthesis, sol-gel processing, aerogel preparation, calcination, and ball milling.
- **Electrode processing:** Battery electrode preparation, slurry formulation, coating, and calendering.
- **Cell fabrication and evaluation:** Coin-cell and pouch-cell fabrication and electrochemical performance evaluation.
- **Microscopy: 60 hours of independent JEOL F200 TEM operation**; hands-on experience with AFM and SEM.
- **Materials characterization:** Interpretation and analysis of Raman spectroscopy, X-ray diffraction (XRD), and BET surface-area data.
- **Equipment calibration:** Laboratory equipment calibration and operating-parameter adjustment.
- **Design and prototyping:** Autodesk Fusion 360, 3D modeling, design for additive manufacturing, custom extruder development, and syringe-based paste-extrusion printing.
- **Printer customization:** Marlin firmware installation and flashing, code/configuration modification, and microstepping adjustments.
- **Research tools:** Origin for scientific plotting; Microsoft Office.

RELEVANT COURSES

1. 3-D Rapid Prototyping and Modeling
2. Practical Approach TEM
3. Electrochemical Energy Storage Systems: Batteries, Fuel Cells, and Supercapacitors
4. Nanoelectronics: Principles and Devices
5. Electronic Motors and Devices
6. Electron Microscopy

B.Tech and M.Tech in Nanotechnology